



Reading the Drift Before the Workaround: Forecasting Exam-Integrity Vulnerability from Capability Trajectories

Capability-drift telemetry as an early-warning signal for assessment-integrity vulnerability, ahead of detection.

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The problem

Assessment-integrity policy updates only after a workaround is reported — months after the enabling tool cleared review. Little is known whether capability trajectory can move that trigger earlier.

Research questions

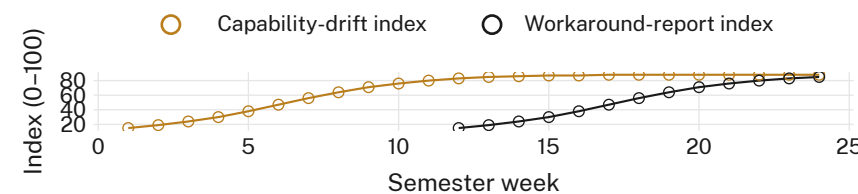
- Does capability drift predict a tool's first misuse?
- Is the lead time stable across cycles and tools?
- Does a forecast trigger beat an incident trigger?

Approach

- $n = 214$ tool updates · 3 assessment cycles · 2024–2026
- Capability-drift index regressed on hand-coded misuse dates
- 14-week forecast horizon pre-registered; no post hoc changes

Findings

The capability-drift index anticipated workaround emergence with a stable 11.4-week lead ($r = 0.79$), holding across all three cycles.



Capability-drift index (gold) crossed threshold in week 8; the workaround-report index (ink) crossed it in week 19 — an 11-week lag, stable across cycles (2024–2026).

Implications

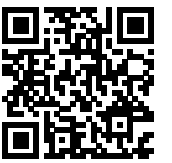
Results are consistent with integrity policy shifting from reactive to anticipatory without a new sanction. The model surprises only on genuinely novel tool classes. Next: extend the index to unreleased tools visible in developer discourse.

Selected references

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De-identified tool-release metadata only; no student submissions retained. Supported by a School of Emergent Priorities seed grant.

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“The workaround arrives 11.4 weeks
after the capability that enables it.”

Capability-drift index vs. workaround-report index, three assessment cycles (2024–2026).